

**SHETH L.U.J AND SIR M.V. COLLEGE**  
**SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU**

**Practical No. 2 (Excel)**

**Aim:** Working with Data Sources and Data Preparation in Tableau

- A. Connect to different data sources (Excel, CSV, Database)
- B. Create and use data extracts vs live connections
- C. Manage metadata (rename, hide, change data types)
- D. Perform joins (inner, left, right) on multiple tables
- E. Apply data blending between two sources
- F. Implement different types of filters (dimension, measure, date)
- G. Optimize performance using extracts and filtering

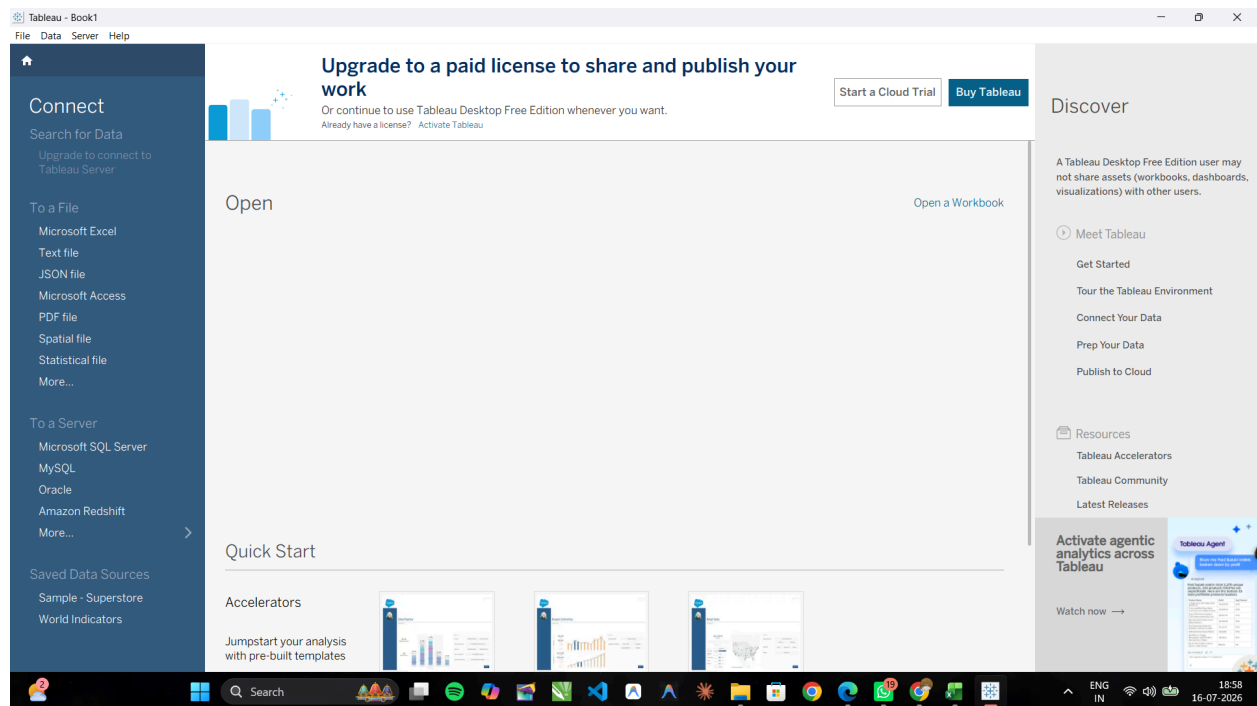
Dataset: Inventory-Records-Sample-Data.xlsx

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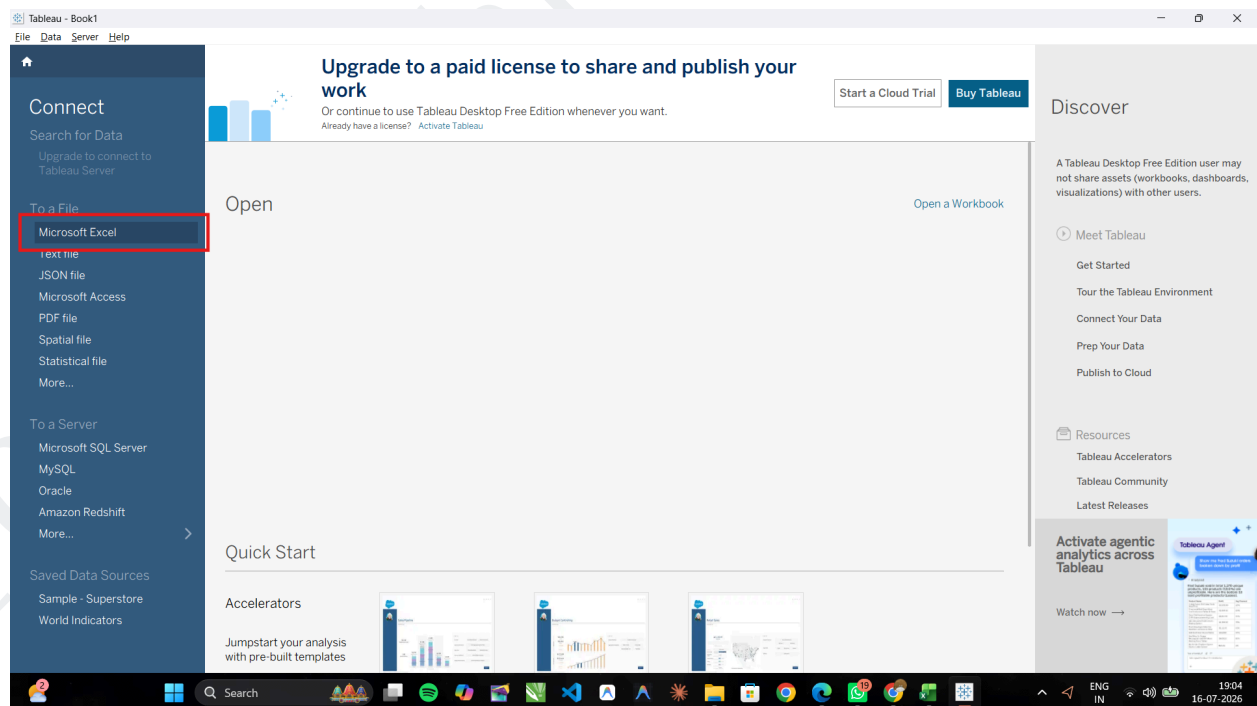
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### Step A: Connect to different data sources (Excel, CSV, Database)

#### 1. Launch Tableau Desktop on your machine



#### 2. Go to **Connect** → **Microsoft Excel**, select **Select Inventory-Records-Sample-Data.xlsx**, and click **Open**.

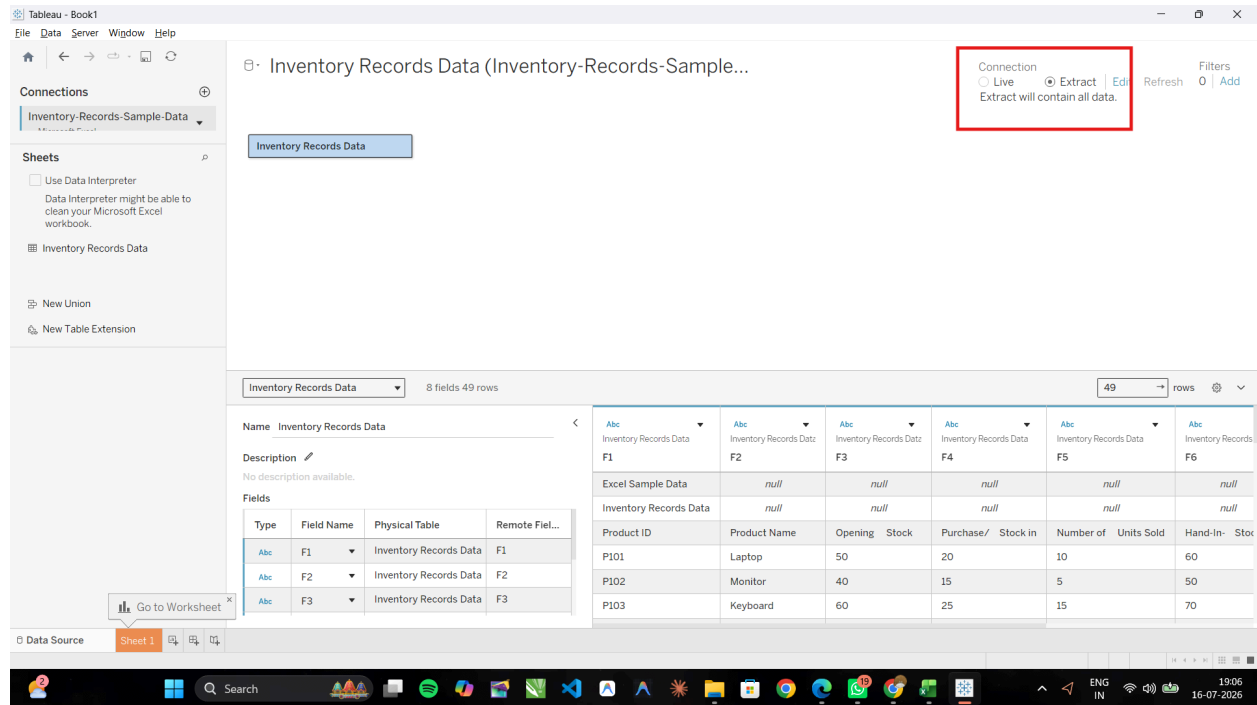


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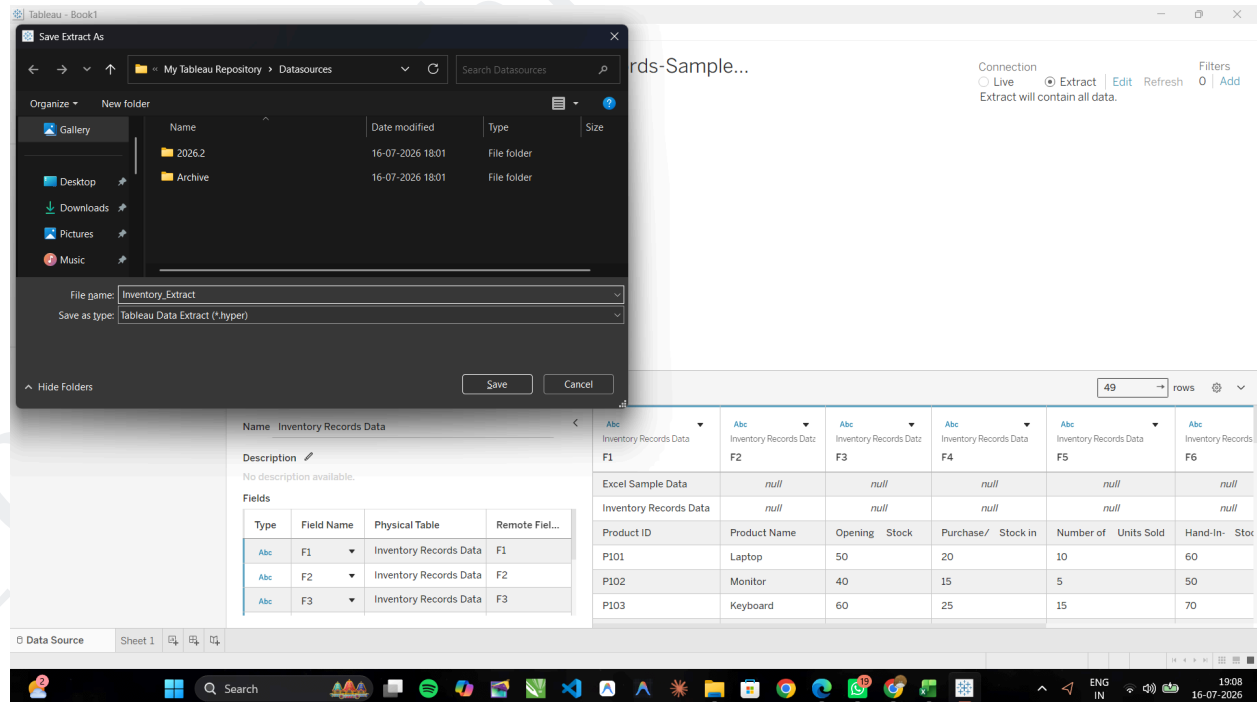
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### Step B: Creating Data Extracts vs. Live Connections

1. In the **Data Source** view, locate the connection mode toggle in the top-right corner and switch the active selection from **Live** to **Extract**.



2. Click **Sheet 1**, name the extract file **Inventory\_Extract**, and click **Save**.

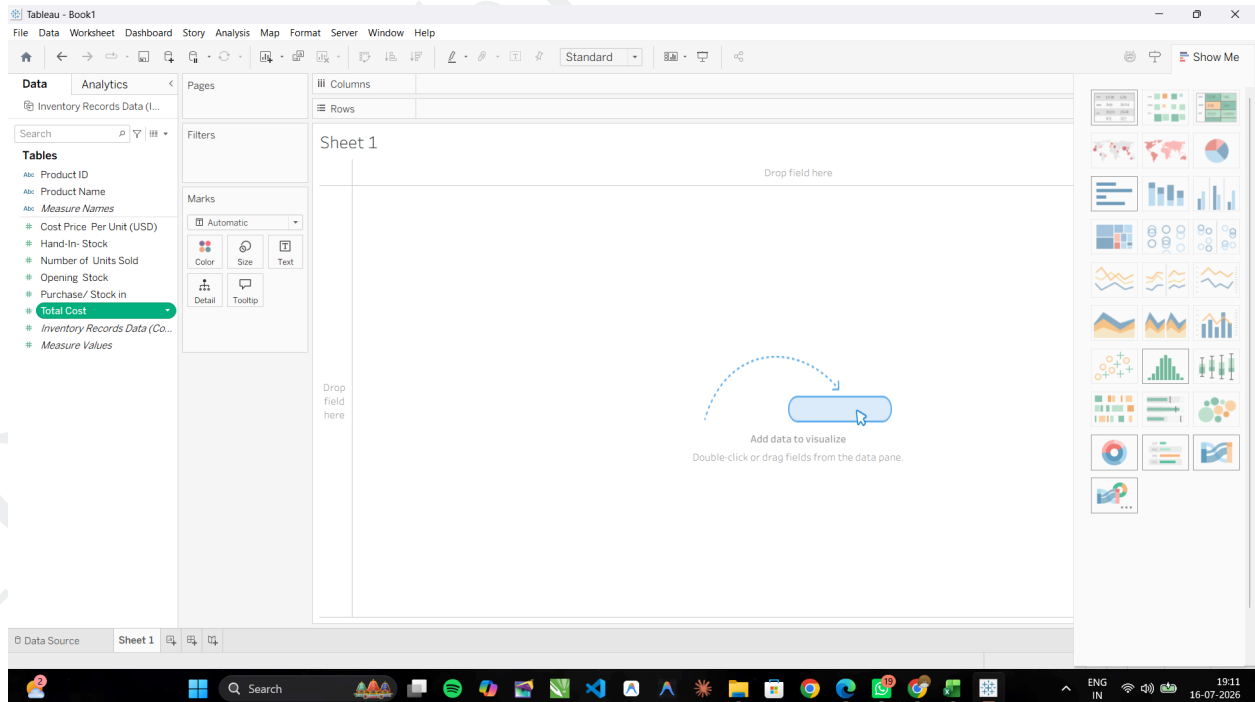
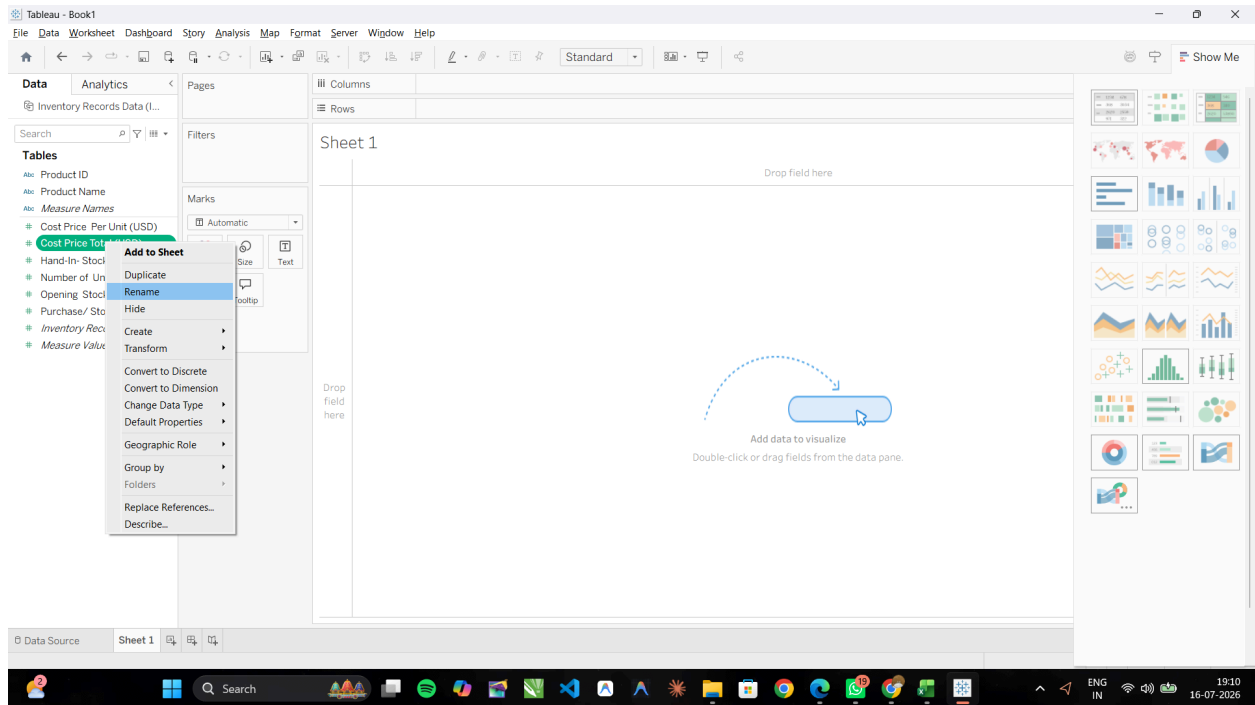


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### Step C: Managing Operational Metadata

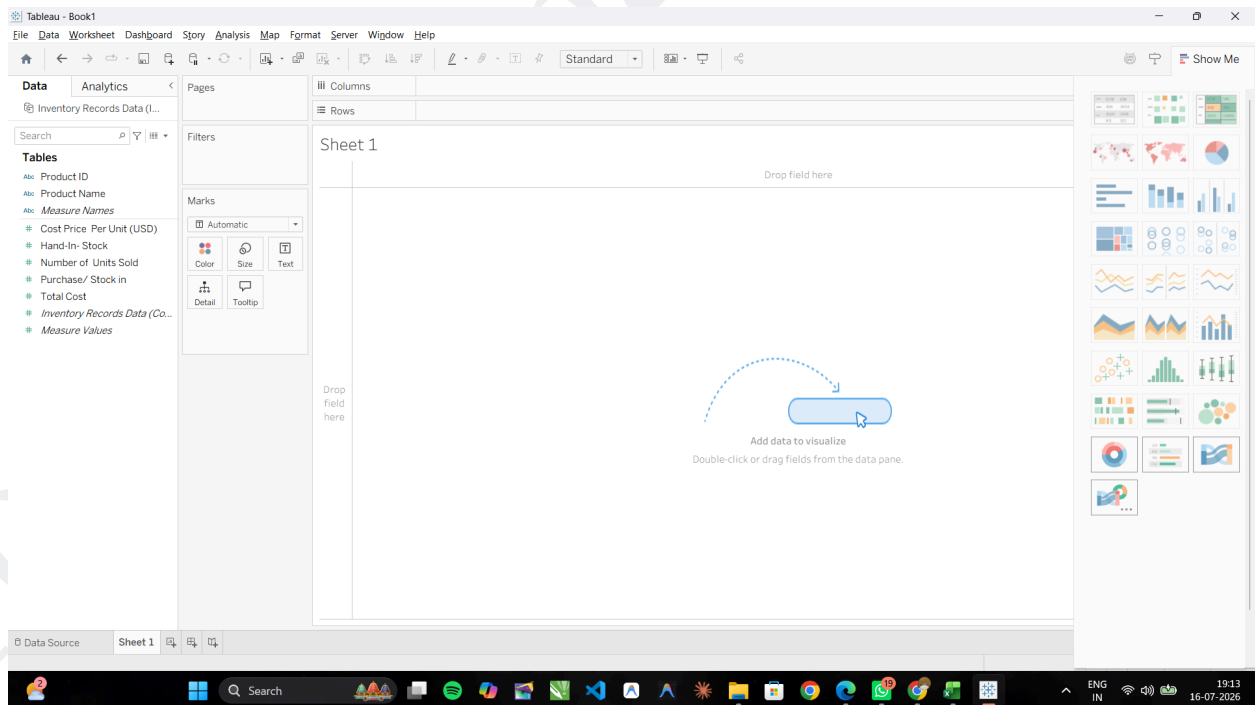
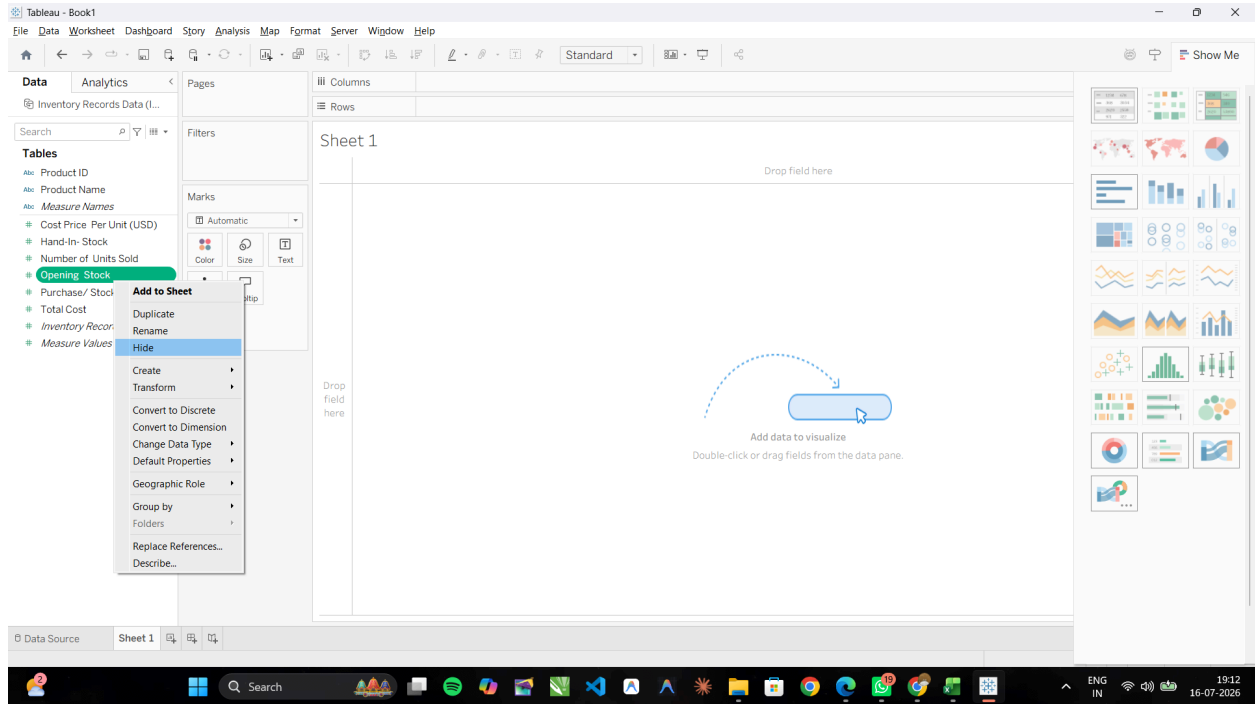
1. Rename a Field: Right-click **Cost Price Total (USD)** and rename it to **Total Cost**.



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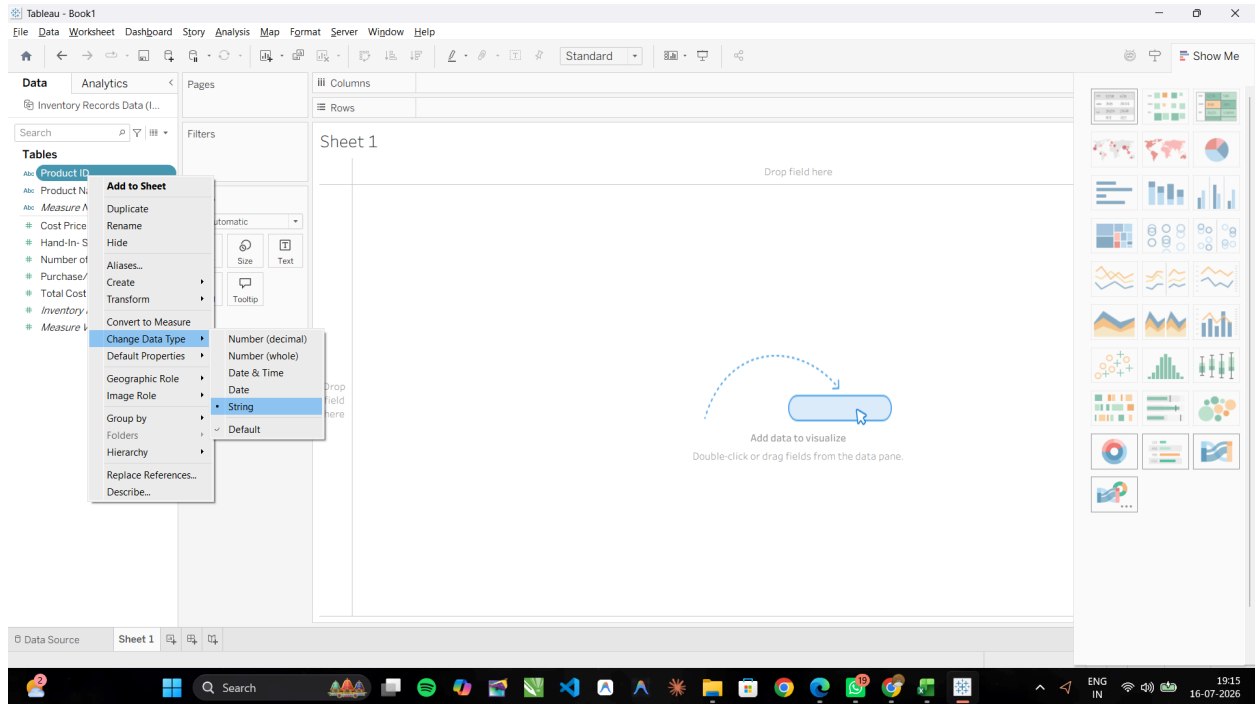
### 2. Hide a Field: Right-click Opening Stock and select Hide.



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3. Change a Data Type: Click the data type icon beside **Product ID** and select **String**.

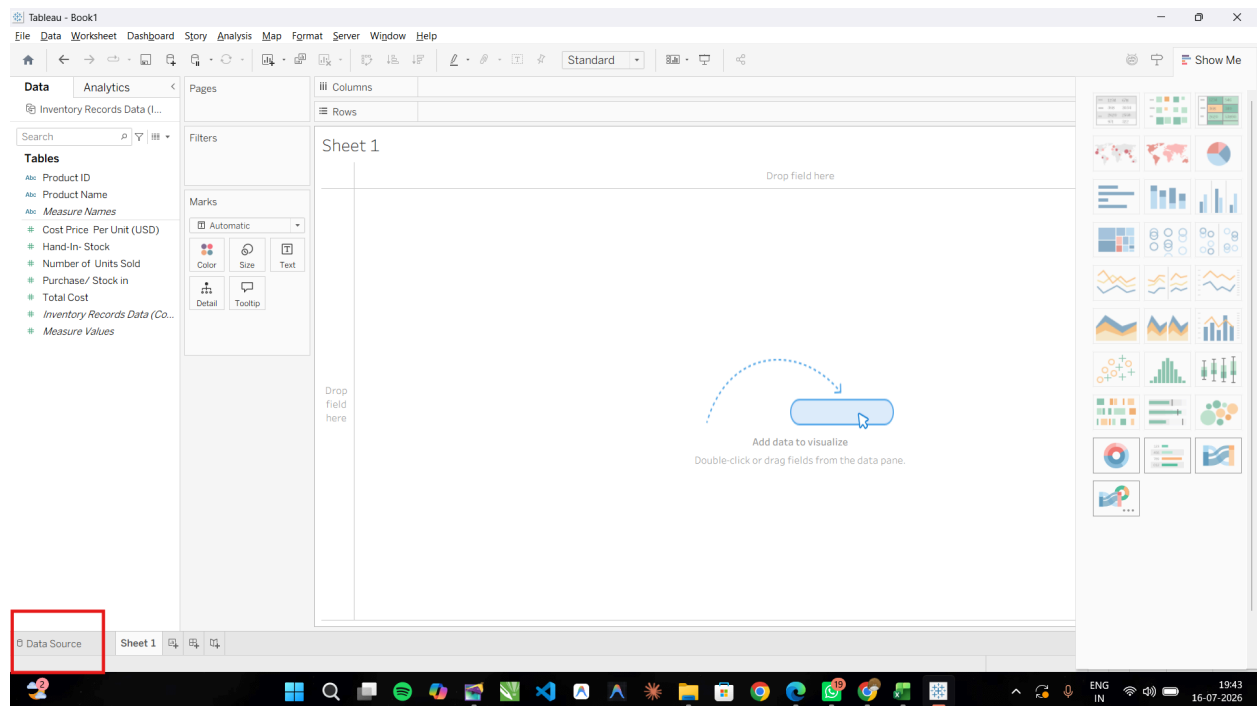


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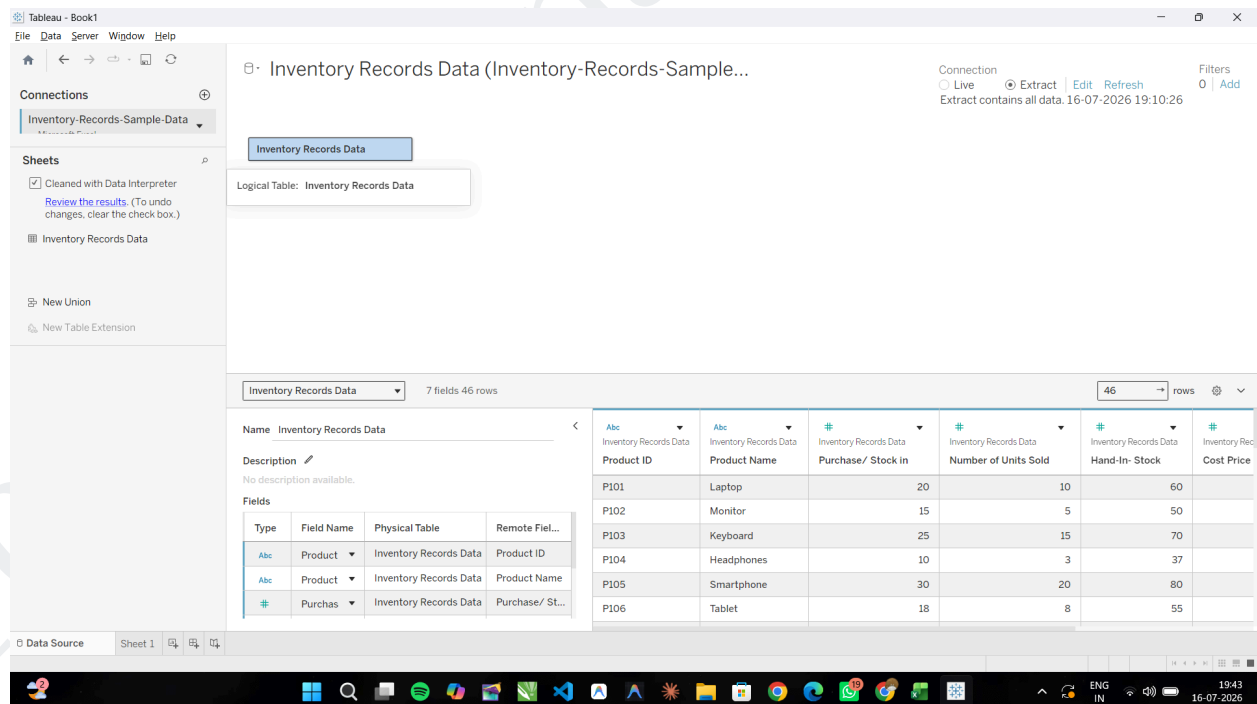
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### Step D: Performing Relational Joins on Multiple Tables

#### 1. Open the Data Source page.



#### 2. Double-click **Inventory Records Data** to enter the Physical Layer.



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### 3. Drag the Sales sheet next to Inventory Records Data.

The screenshot shows the Tableau interface with the 'Inventory Records Data' data source selected. The 'Sales' sheet is being added to the data source. The 'Fields' section shows the 'Sales' sheet with fields: Product ID (Sales), Number of Units Sold, and Cost Price Per Unit (USD). The 'Data Source' section shows the 'Sales' sheet with 3 fields and 15 rows.

| Product ID (Sales) | Number of Units Sold | Cost Price Per Unit (USD) |
|--------------------|----------------------|---------------------------|
| P001               | 45                   | 25.5000                   |
| P002               | 30                   | 18.0000                   |
| P003               | 62                   | 12.7500                   |
| P004               | 15                   | 45.0000                   |
| P005               | 28                   | 33.5000                   |
| P006               | 50                   | 20.0000                   |

### 4. Click the Join (Venn Diagram) icon.

The screenshot shows the Tableau interface with the 'Inventory Records Data' data source selected. The 'Sales' sheet is being added to the data source. The 'Fields' section shows the 'Sales' sheet with fields: Product ID, Product Name, Purchase/ Stock in, Number of Units Sold, Hand-In- Stock, and Cost Price F. The 'Data Source' section shows the 'Inventory Records Data' sheet with 10 fields and 0 rows.

| Product ID | Product Name | Purchase/ Stock in | Number of Units Sold | Hand-In- Stock | Cost Price F |
|------------|--------------|--------------------|----------------------|----------------|--------------|
|------------|--------------|--------------------|----------------------|----------------|--------------|



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5. Set the join condition as: **Product ID = Product ID**

Observe the data preview using: **Inner Join, Left Join**

The screenshot shows the Tableau interface with the 'Inventory Records Data' and 'Sales1' tables selected for a join. The join type is set to 'Inner'. The join condition is 'Product ID = Product ID'. The data preview shows 10 fields and 0 rows.

Inventory Records Data is made of 2 tables.

Inventory Records Data Sales1

Join

| Join                             | Left                  | Right                 | Full Outer            |
|----------------------------------|-----------------------|-----------------------|-----------------------|
| <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Data Source: Sales1

| Product ID          | = | Product ID ... |
|---------------------|---|----------------|
| Add new join clause |   |                |

Inventory Records Data

10 fields 0 rows

| Name        | Inventory Records Data    |               |          |
|-------------|---------------------------|---------------|----------|
| Description | No description available. |               |          |
| Fields      |                           |               |          |
| Type        | Field Name                | Physical T... | Rem...   |
| Abc         | Product ID                | Inventory ... | Produ... |
| Abc         | Product Name              | Inventory ... | Produ... |
| #           | Purchase/ Stock in        | Inventory ... | Purch... |

| Abc                    | Abc                    | #                      | #                      | #                      | #              |
|------------------------|------------------------|------------------------|------------------------|------------------------|----------------|
| Inventory Records Data | Inventory Records Data | Inventory Records Data | Inventory Records Data | Inventory Records Data | Inventory Reco |
| Product ID             | Product Name           | Purchase/ Stock in     | Number of Units Sold   | Hand-In- Stock         | Cost Price F   |

6. Leave the join configuration set to **Left Join** for the remainder of the exercise.

The screenshot shows the Tableau interface with the 'Inventory Records Data' and 'Sales1' tables selected for a join. The join type is set to 'Left'. The join condition is 'Product ID = Product ID'. The data preview shows 46 rows.

Inventory Records Data is made of 2 tables.

Inventory Records Data Sales1

Join

| Join                  | Left                             | Right                 | Full Outer            |
|-----------------------|----------------------------------|-----------------------|-----------------------|
| <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Data Source: Sales1

| Product ID          | = | Product ID ... |
|---------------------|---|----------------|
| Add new join clause |   |                |

Inventory Records Data

10 fields 46 rows

| Name        | Inventory Records Data    |               |          |
|-------------|---------------------------|---------------|----------|
| Description | No description available. |               |          |
| Fields      |                           |               |          |
| Type        | Field Name                | Physical T... | Rem...   |
| Abc         | Product ID                | Inventory ... | Produ... |
| Abc         | Product Name              | Inventory ... | Produ... |
| #           | Purchase/ Stock in        | Inventory ... | Purch... |

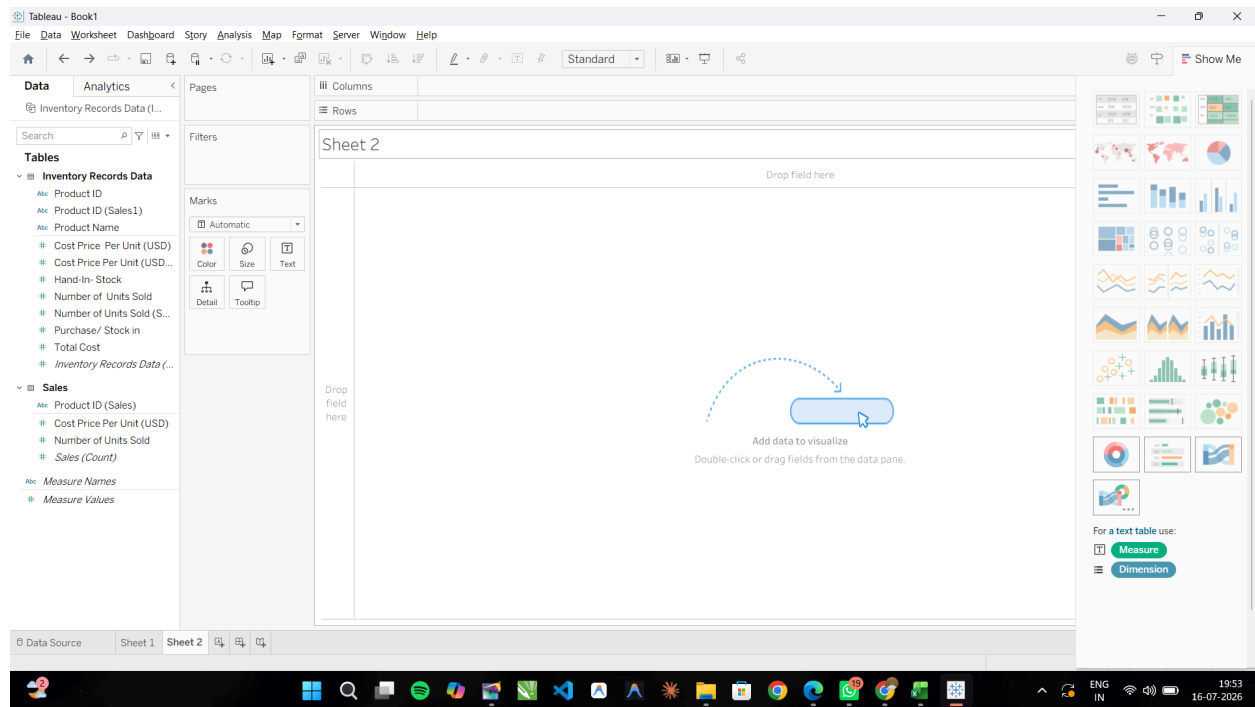
| Abc                    | Abc                    | #                      | #                      | #                      | #              |
|------------------------|------------------------|------------------------|------------------------|------------------------|----------------|
| Inventory Records Data | Inventory Records Data | Inventory Records Data | Inventory Records Data | Inventory Records Data | Inventory Reco |
| Product ID             | Product Name           | Purchase/ Stock in     | Number of Units Sold   | Hand-In- Stock         | Cost Price     |
| P101                   | Laptop                 | 20                     | 10                     | 60                     |                |
| P102                   | Monitor                | 15                     | 5                      | 50                     |                |
| P103                   | Keyboard               | 25                     | 15                     | 70                     |                |
| P104                   | Headphones             | 10                     | 3                      | 37                     |                |
| P105                   | Smartphone             | 30                     | 20                     | 80                     |                |
| P106                   | Tablet                 | 18                     | 8                      | 55                     |                |

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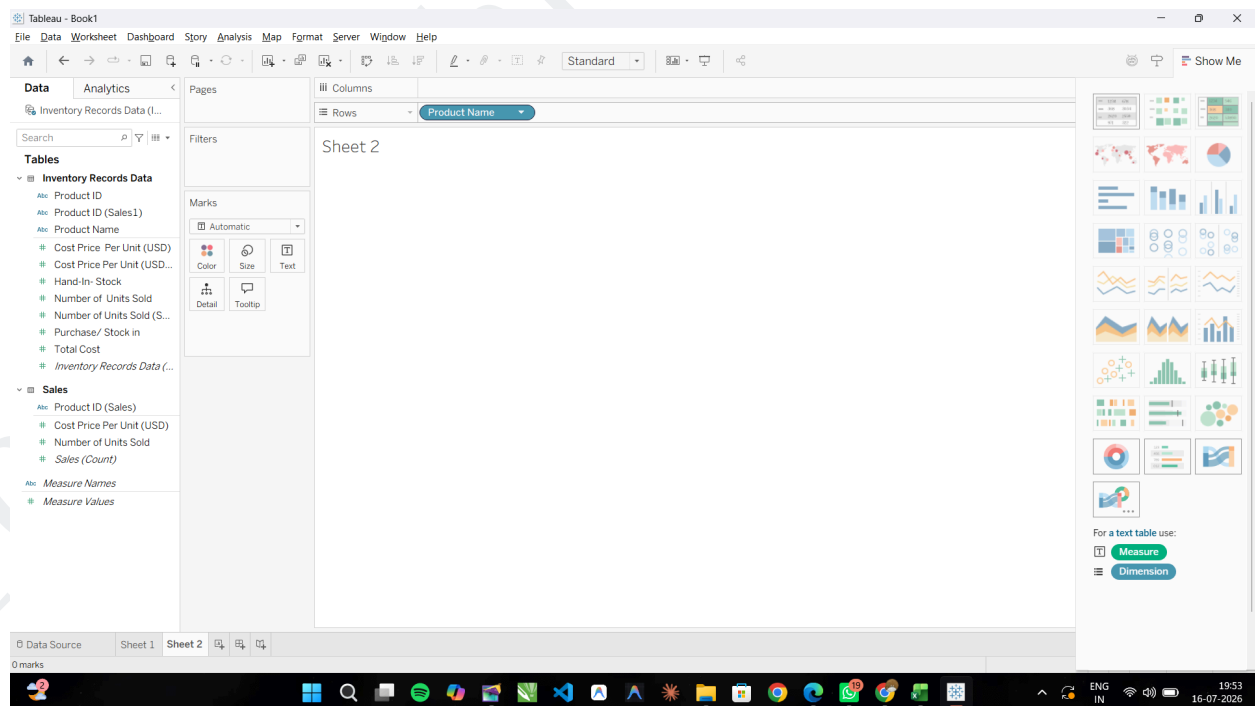
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#### Step E: Applying Data Blending Between Independent Sources

1. Create a new worksheet by clicking the **New Worksheet (+)** icon to open **Sheet 2**. Ensure **Inventory Records Data** is selected as the **Primary Data Source**.



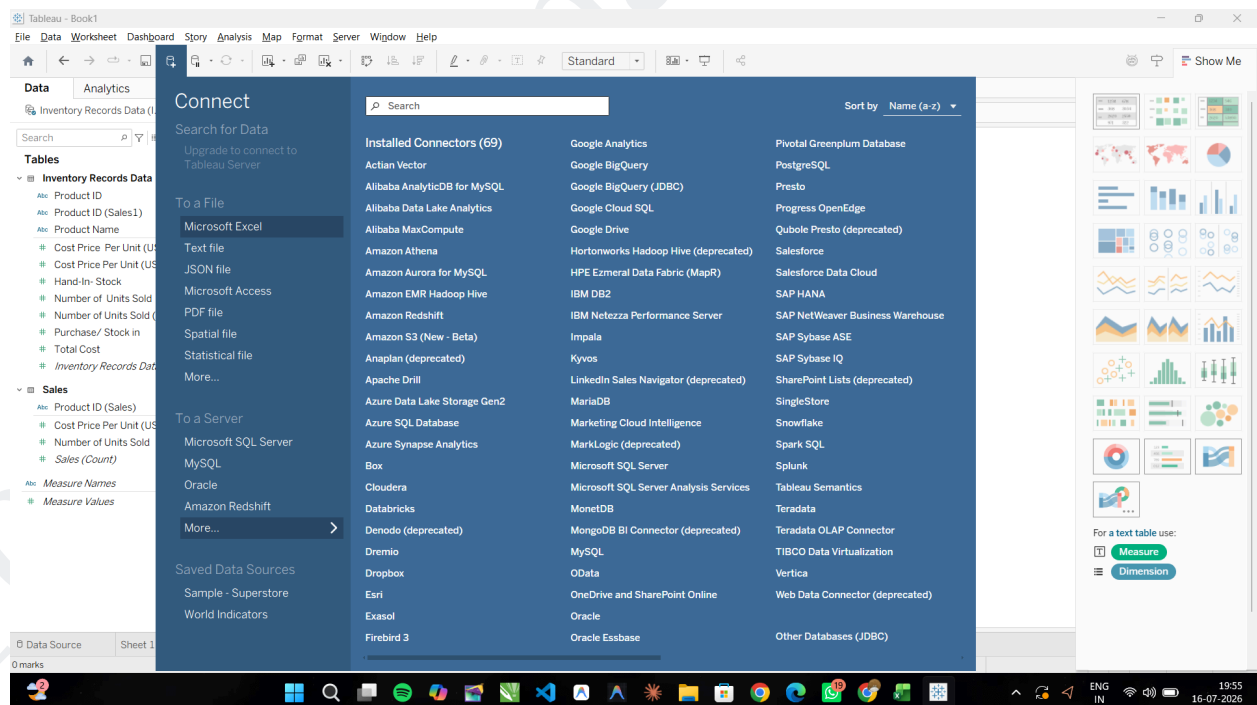
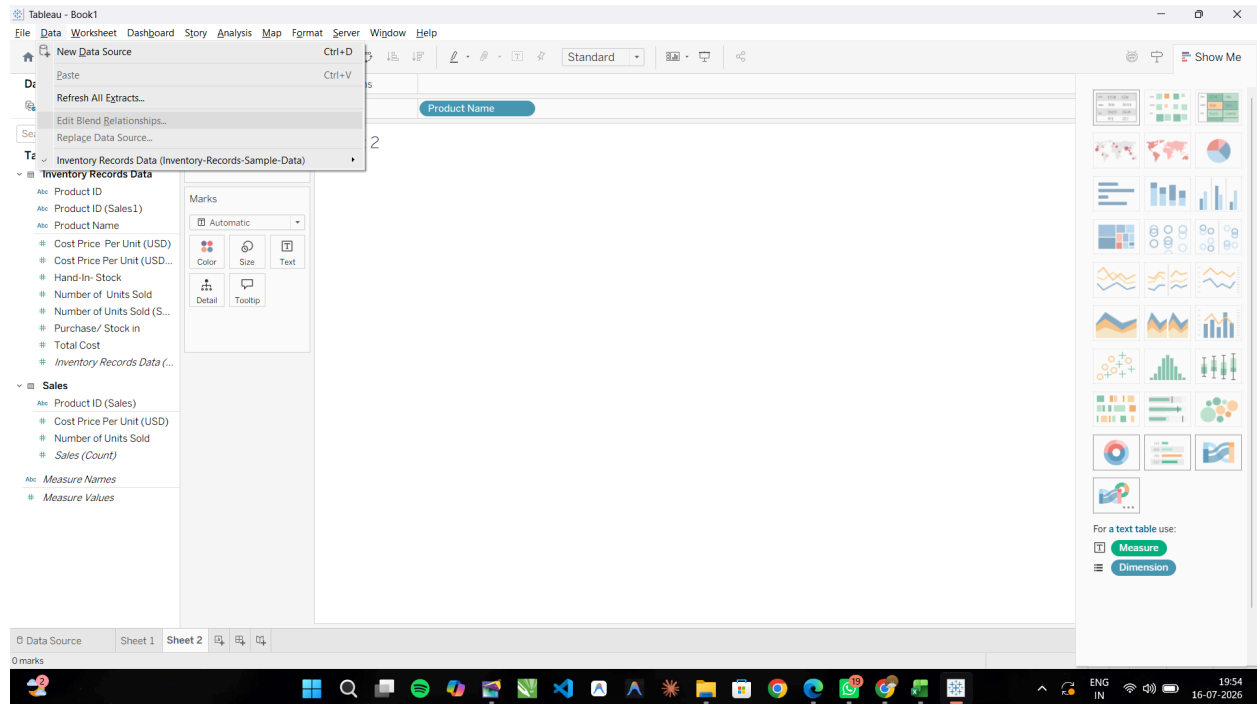
2. Drag **Product Name** to the **Rows** shelf.



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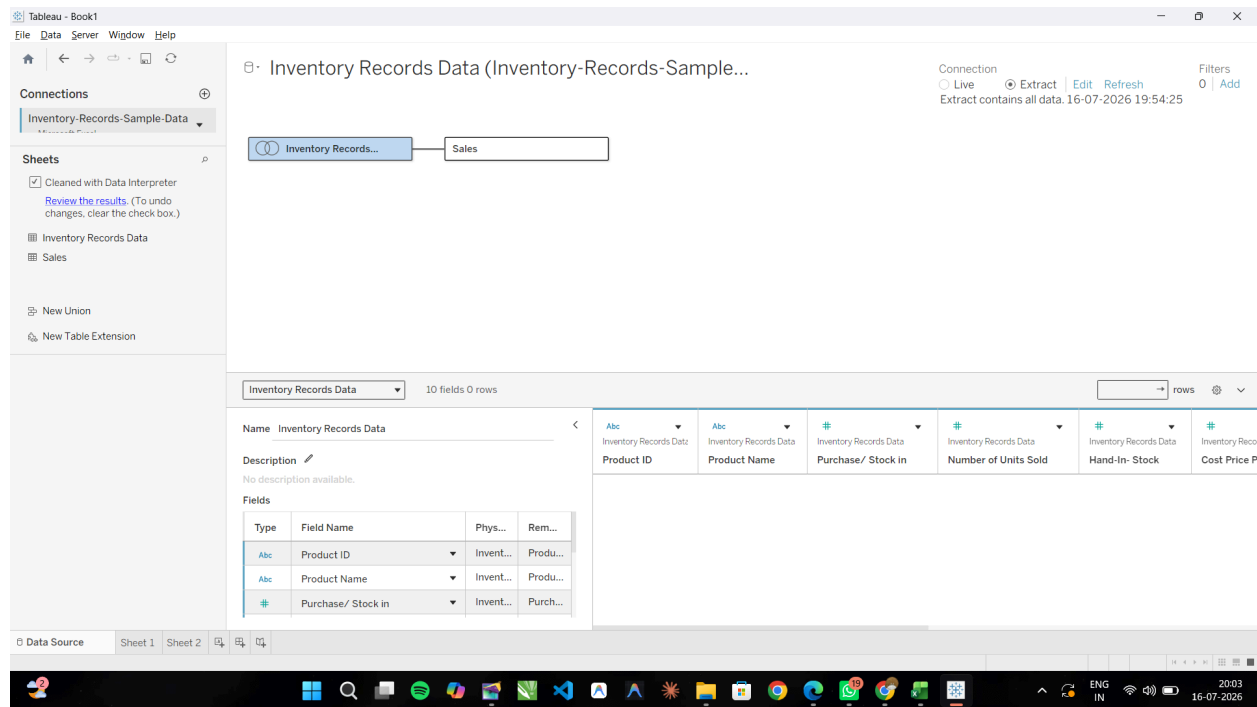
3. Go to **Data** → **New Data Source** and connect to the same Excel file again.



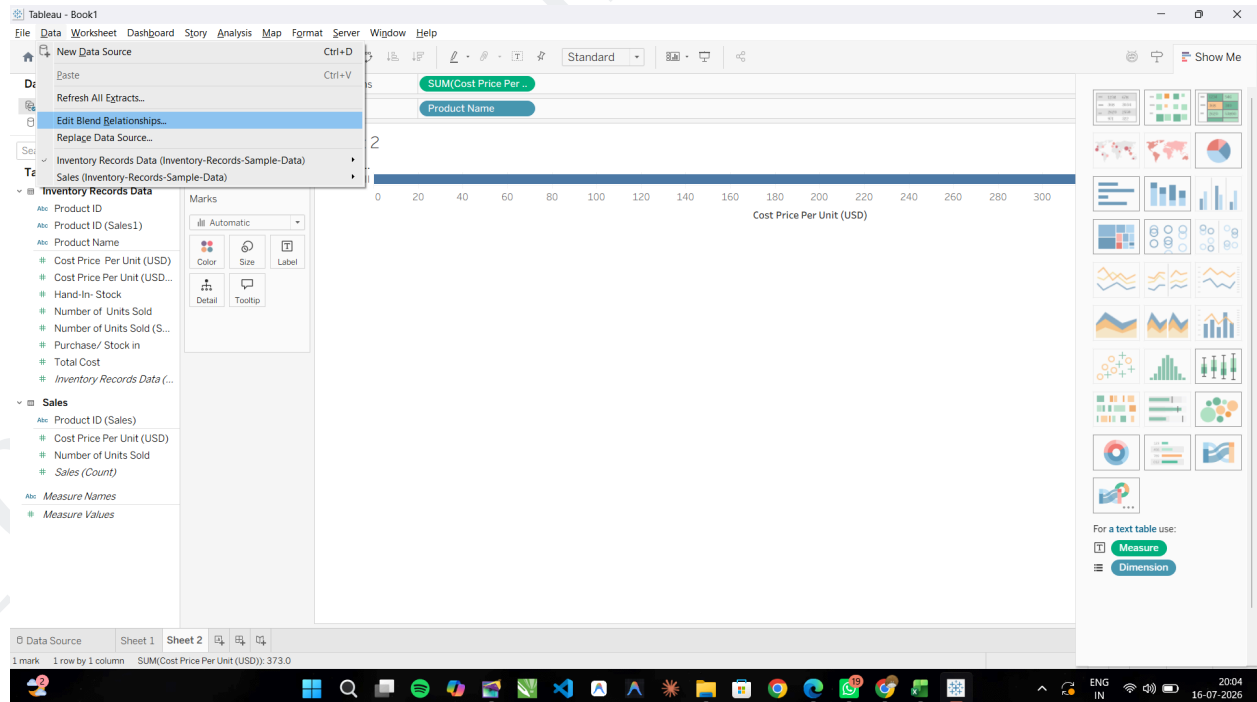
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### 4. Select the Sales sheet as the Secondary Data Source.



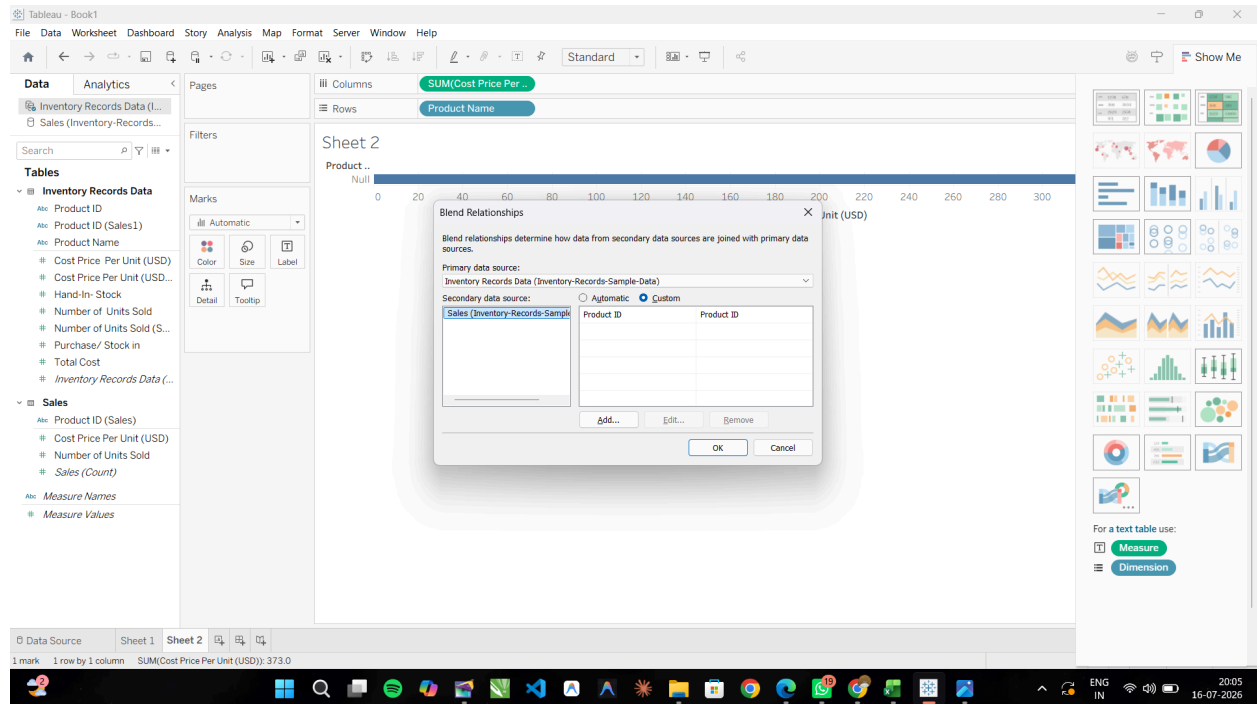
### 5. Go to Data → Edit Blend Relationships.



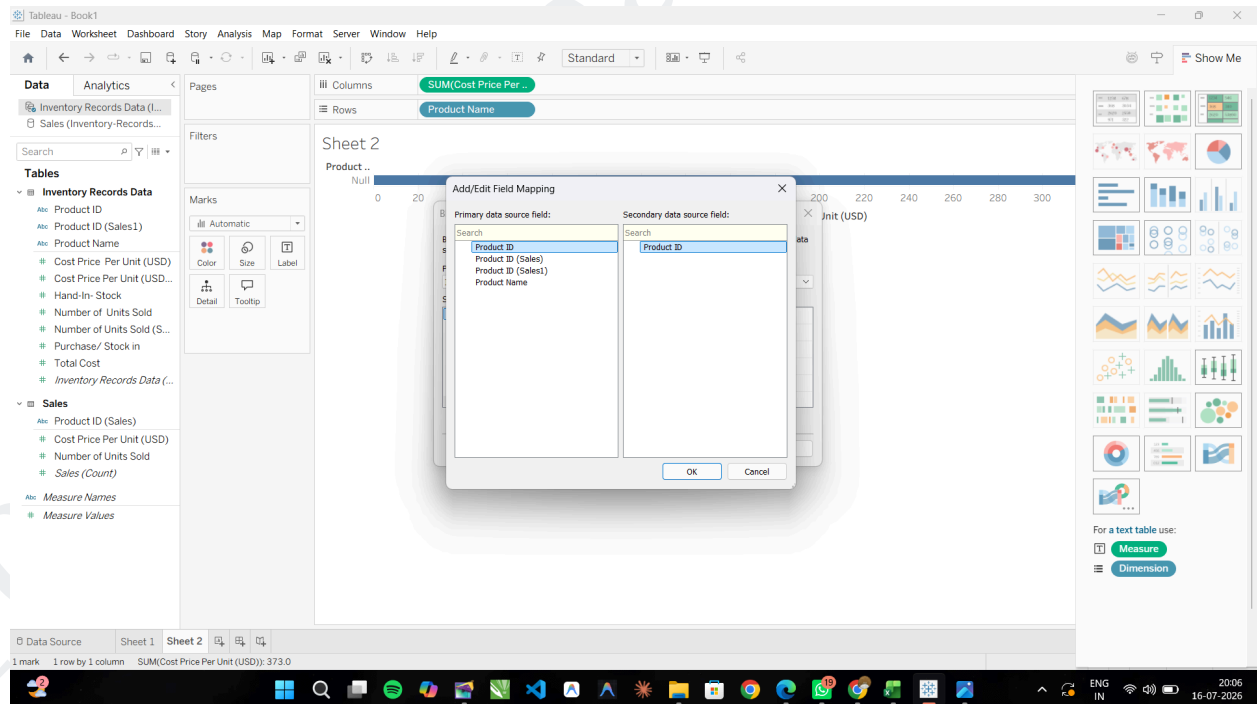
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#### 6. Select **Custom**, click **Add**, and map: **Product ID = Product ID**



#### 7. Click **OK** to complete the Data Blending relationship.

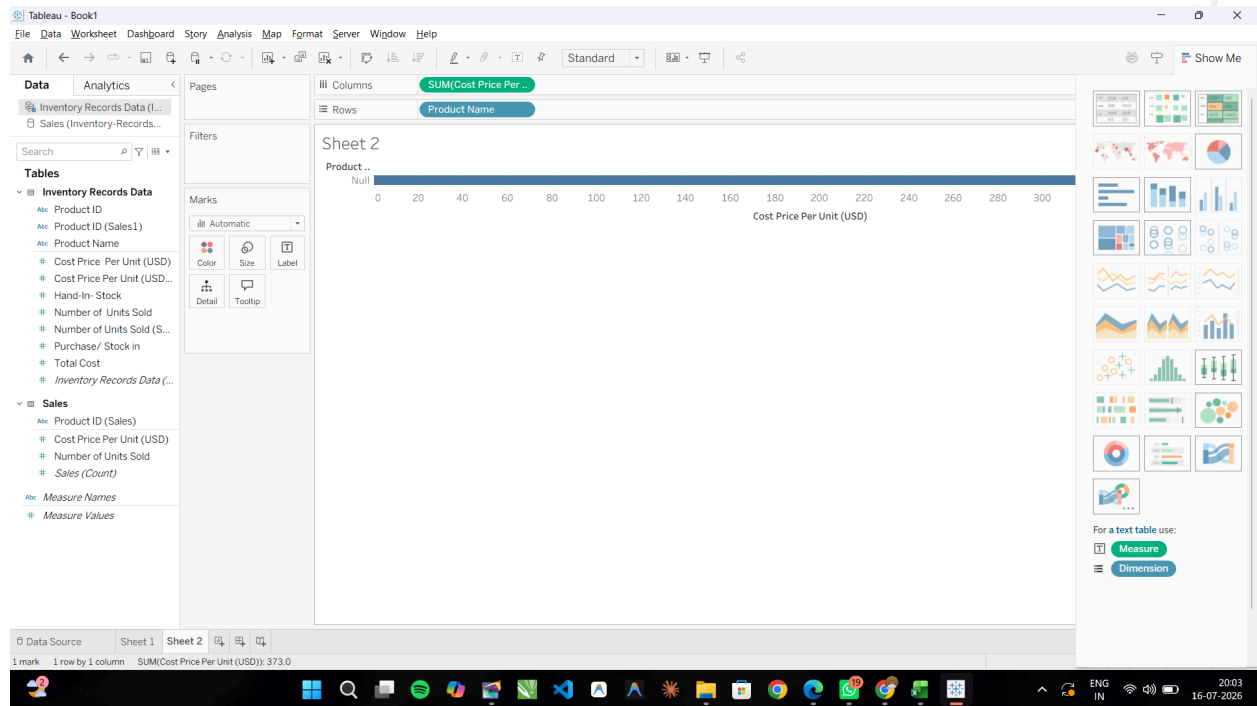


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#### Step F: Implementing Different Types of Filters

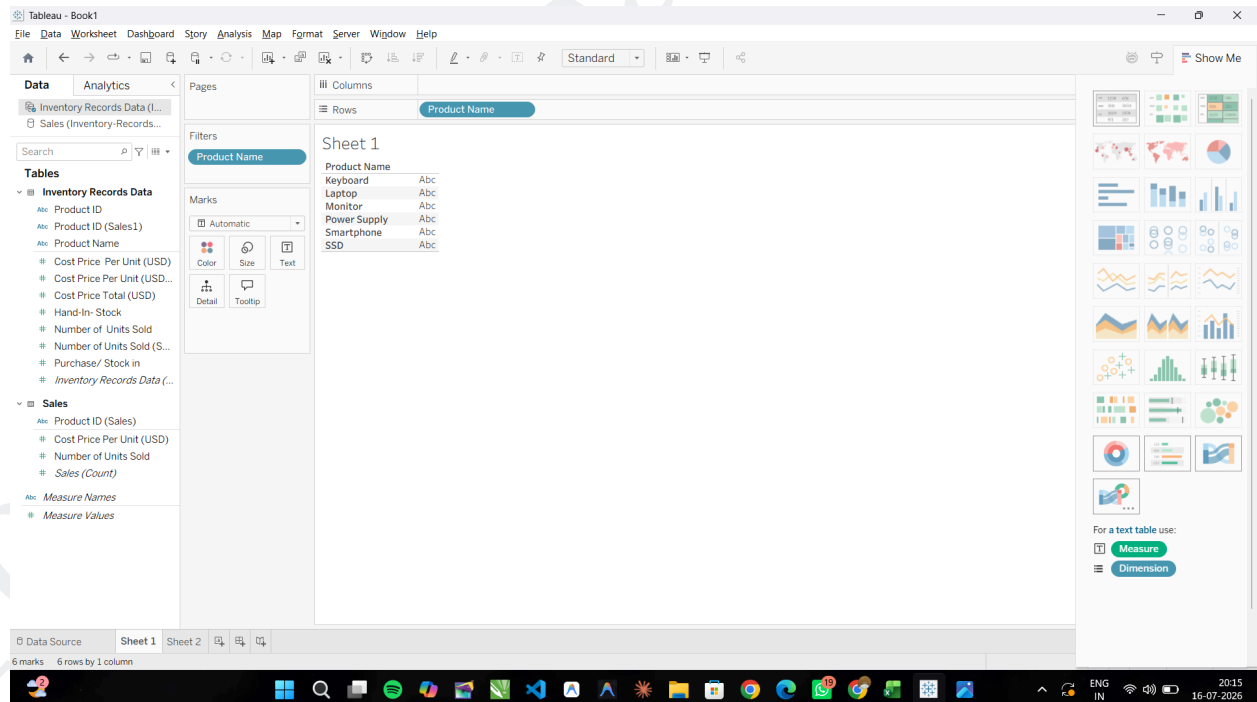
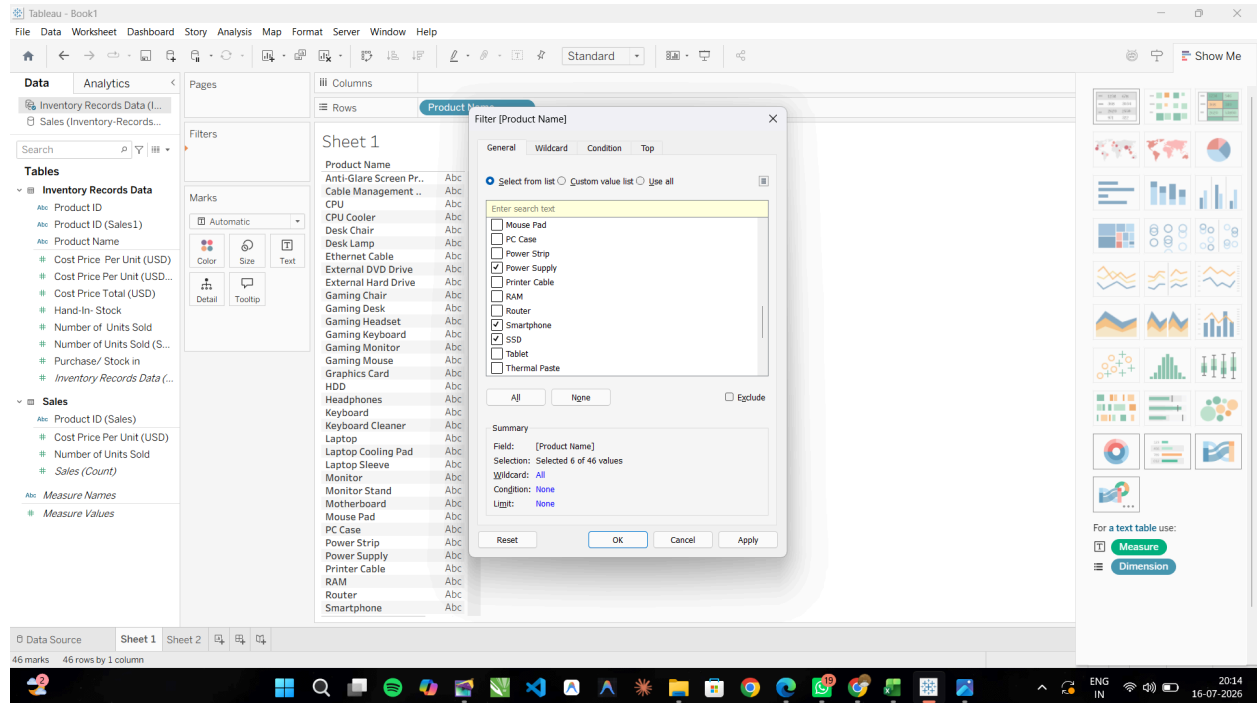
1. Open a **new worksheet**. Drag **Number of Units Sold** to the **Columns shelf** and **Product Name** to the **Rows shelf**.



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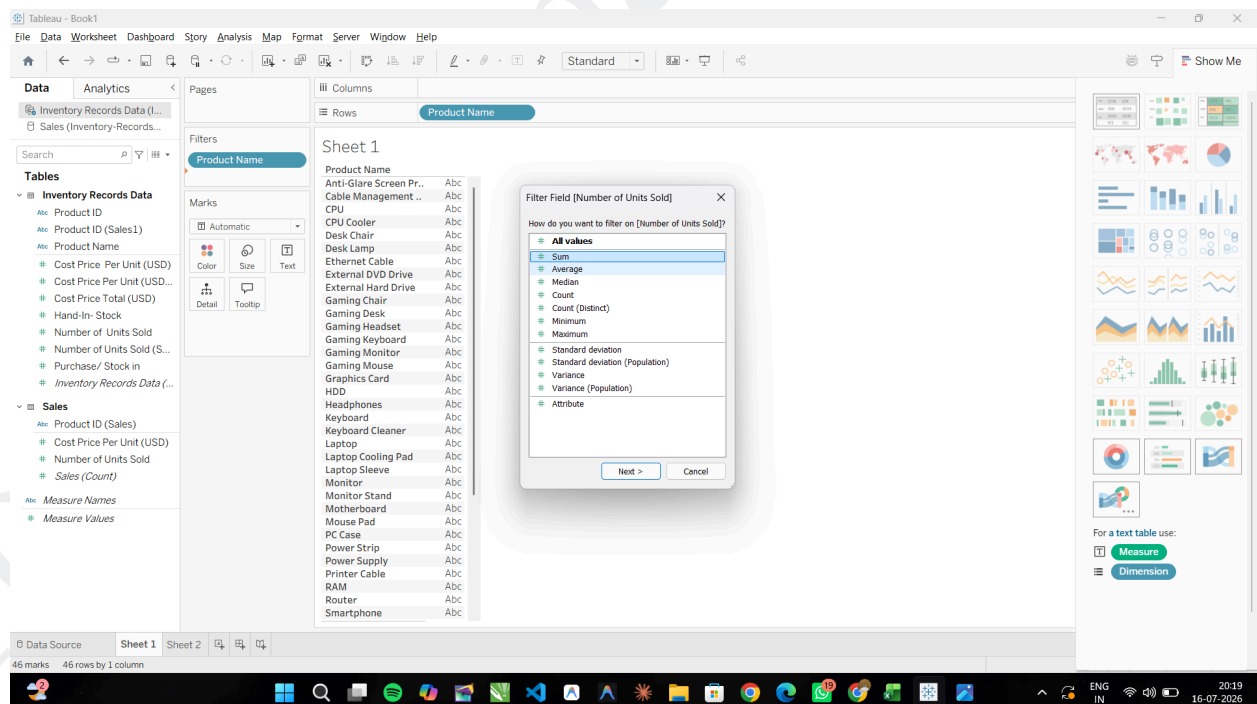
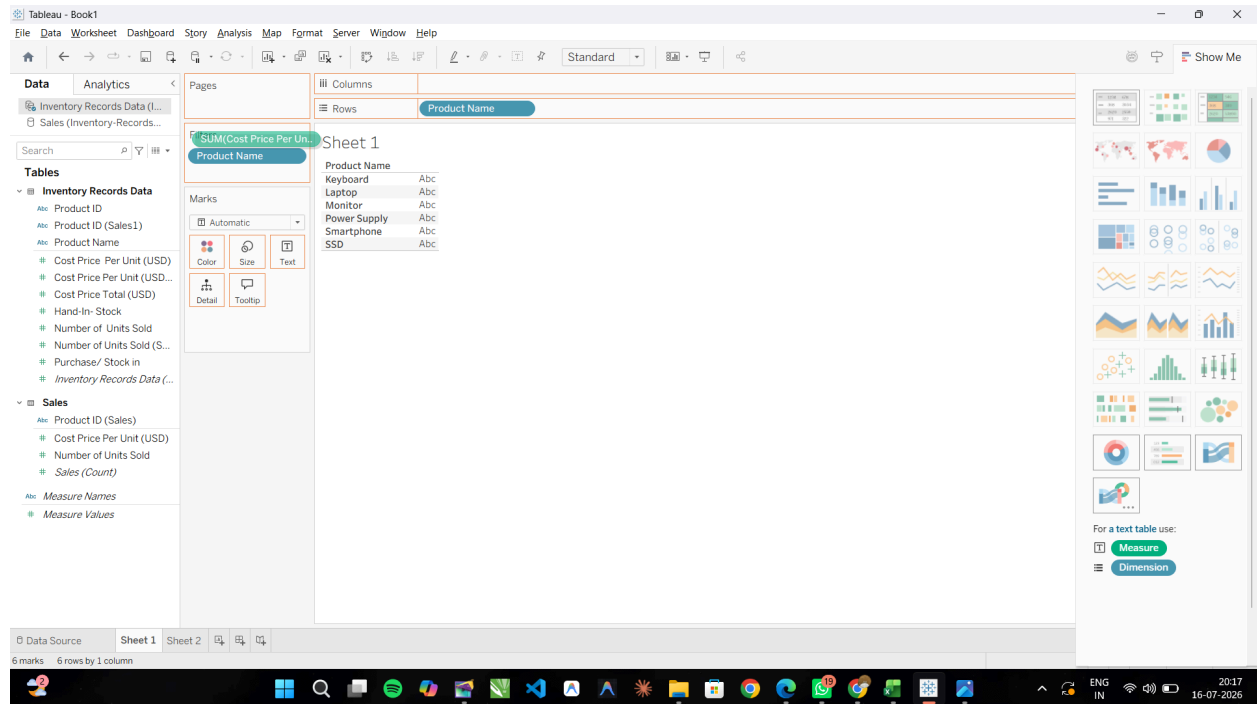
**2. Dimension Filter:** Drag **Product Name** to the **Filters** shelf, select the required products (e.g., Laptop, Monitor, Keyboard, Smartphone), and click **OK**.



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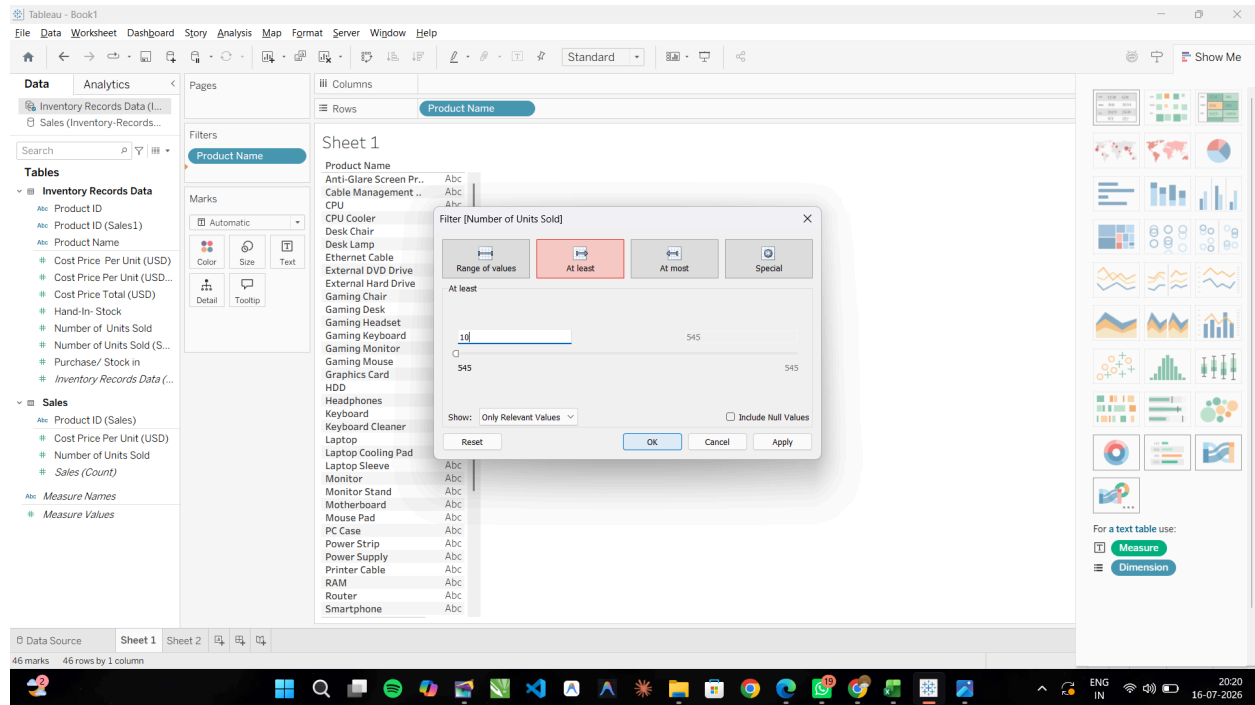
**3. Measure Filter:** Drag **Number of Units Sold** to the **Filters** shelf, select **All Values**, click **Next**, choose **At Least**, enter **10**, and click **OK**.



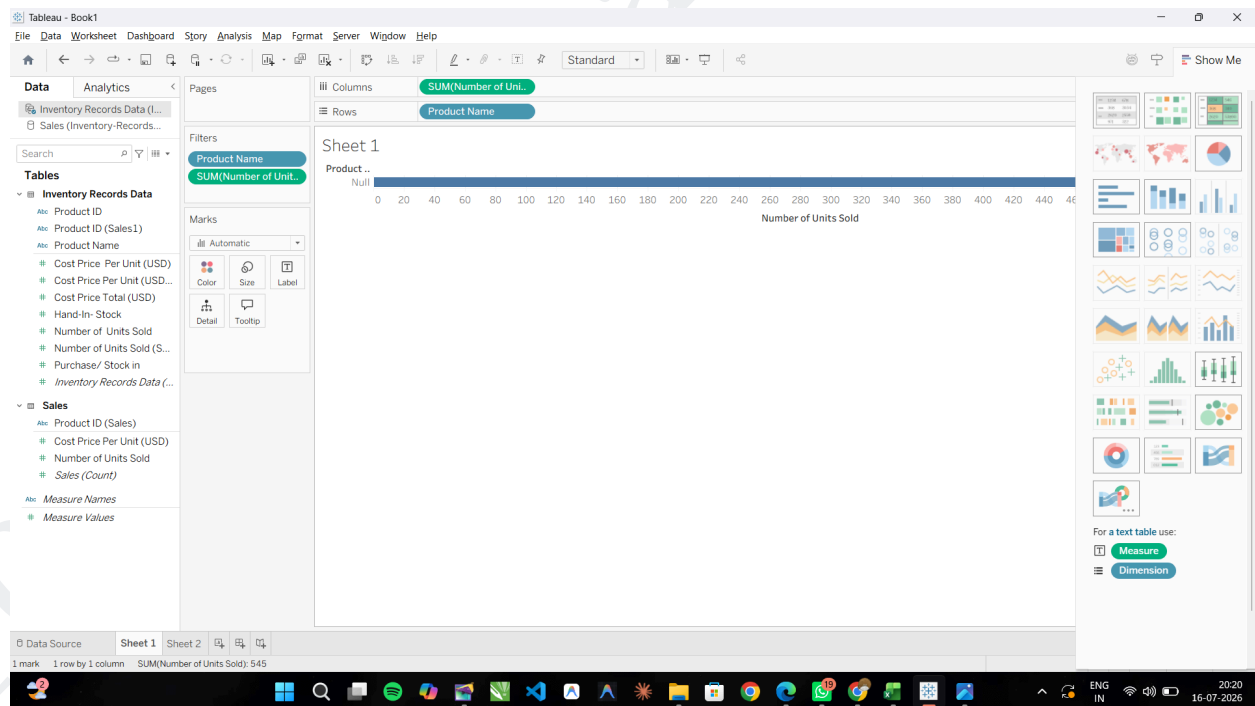


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**4. Date Filter:** The current Inventory dataset does not contain a **Date** field. Therefore, a Date Filter cannot be applied.



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### Step G: Optimizing Dashboard Performance

#### 1. Return to the **Data Source** page.

Tableau - Book1

File Data Server Window Help

Connections

Inventory-Records-Sample-Data

Sheets

Cleaned with Data Interpreter  
[Review the results.](#) (To undo changes, clear the check box.)

Inventory Records Data

Sales

New Union

New Table Extension

Inventory Records Data (Inventory-Records-Sample...)

Connection: Live Extract Edit Refresh  
Extract contains all data. 16-07-2026 20:16:53

Filters: 0 Add

Inventory Records... Sales

Sales 3 fields 15 rows 15 rows

Name Sales

Description No description available.

Fields

| Type | Field Name                | Phys... | Rem...    |
|------|---------------------------|---------|-----------|
| Abc  | Product ID (Sales)        | Sales   | Produ...  |
| #    | Number of Units Sold      | Sales   | Numb...   |
| #    | Cost Price Per Unit (USD) | Sales   | Cost P... |

| Product ID (Sales) | Number of Units Sold | Cost Price Per Unit (USD) |
|--------------------|----------------------|---------------------------|
| P001               | 45                   | 25.5000                   |
| P002               | 30                   | 18.0000                   |
| P003               | 62                   | 12.7500                   |
| P004               | 15                   | 45.0000                   |
| P005               | 28                   | 33.5000                   |
| P006               | 50                   | 20.0000                   |

Data Source Sheet 1 Sheet 2

2022 16-07-2026

#### 2. Under the **Extract** option, click **Edit**, then click **Add**.

Tableau - Book1

File Data Server Window Help

Connections

Inventory-Records-Sample-Data

Sheets

Cleaned with Data Interpreter  
[Review the results.](#) (To undo changes, clear the check box.)

Inventory Records Data

Sales

New Union

New Table Extension

Inventory Records Data (Inventory-Records-Sample...)

Connection: Live Extract Edit Refresh  
Extract contains all data. 16-07-2026 20:16:53

Filters: 0 Add

Inventory Records... Sales

Sales 3 fields 15 rows 15 rows

Name Sales

Description No description available.

Fields

| Type | Field Name                | Phys... | Rem...    |
|------|---------------------------|---------|-----------|
| Abc  | Product ID (Sales)        | Sales   | Produ...  |
| #    | Number of Units Sold      | Sales   | Numb...   |
| #    | Cost Price Per Unit (USD) | Sales   | Cost P... |

| Product ID (Sales) | Number of Units Sold | Cost Price Per Unit (USD) |
|--------------------|----------------------|---------------------------|
| P001               | 45                   | 25.5000                   |
| P002               | 30                   | 18.0000                   |
| P003               | 62                   | 12.7500                   |
| P004               | 15                   | 45.0000                   |
| P005               | 28                   | 33.5000                   |
| P006               | 50                   | 20.0000                   |

Edit Data Source Filters

Filter Scope Details

Add... Edit... Remove

OK Cancel

Data Source Sheet 1 Sheet 2

2022 16-07-2026

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3. Select **Number of Units Sold**, choose **At Least**, enter **10**, and click **OK**.

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane shows 'Inventory-Records-Sample-Data' and 'Sales'. The 'Sheets' pane shows 'Inventory Records Data' and 'Sales'. The main view displays the 'Inventory Records Data' worksheet. A dialog box titled 'Filter [Number of Units Sold]' is open, showing the 'At least' filter type with the value '10' entered. The background worksheet shows the following data:

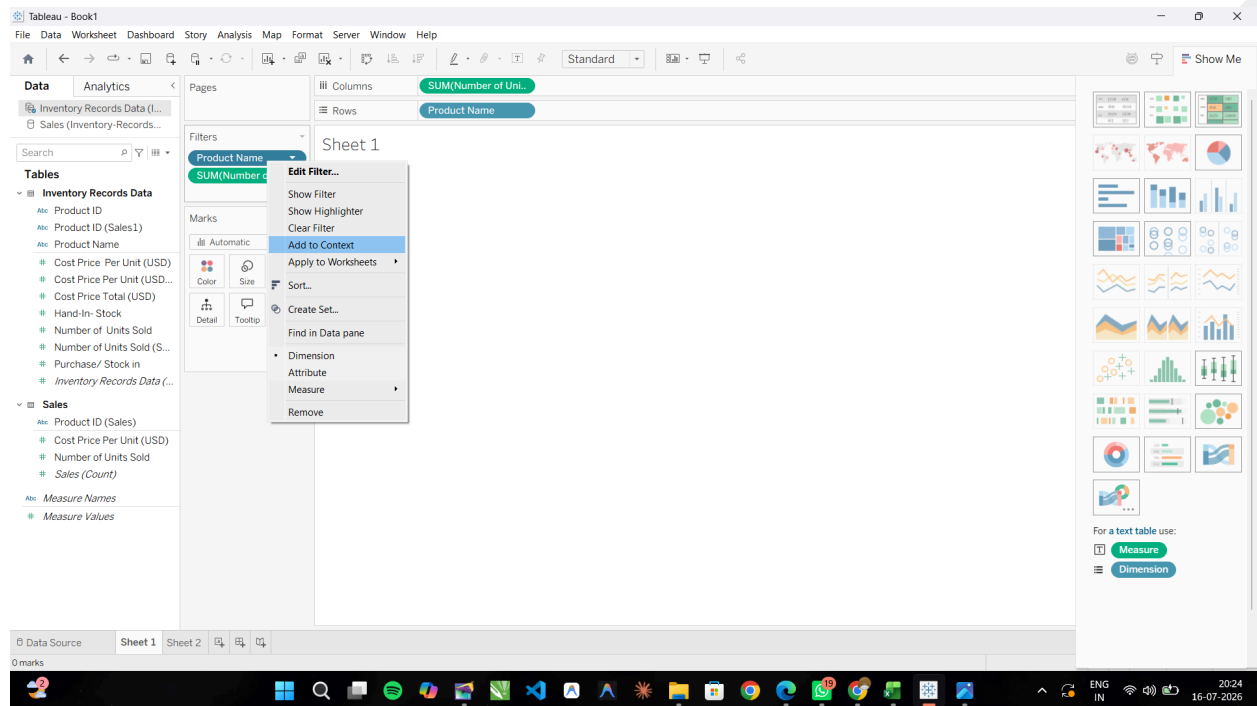
| Product ID (Sales) | Number of Units Sold | Cost Price Per Unit (USD) |
|--------------------|----------------------|---------------------------|
| P001               | 45                   | 25.5000                   |
| P002               | 30                   | 18.0000                   |
| P003               | 62                   | 12.7500                   |
| P004               | 15                   | 45.0000                   |
| P005               | 28                   | 33.5000                   |
| P006               | 50                   | 20.0000                   |

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane shows 'Inventory-Records-Sample-Data' and 'Sales'. The 'Sheets' pane shows 'Inventory Records Data' and 'Sales'. The main view displays the 'Inventory Records Data' worksheet. A dialog box titled 'Edit Data Source Filters' is open, showing the filter 'Number of Units Sold' with the scope 'Inventory Records Data' and the details 'includes values greater than or equal to 10'. The background worksheet shows the following data:

| Product ID (Sales) | Number of Units Sold | Cost Price Per Unit (USD) |
|--------------------|----------------------|---------------------------|
| P001               | 45                   | 25.5000                   |
| P002               | 30                   | 18.0000                   |
| P003               | 62                   | 12.7500                   |
| P004               | 15                   | 45.0000                   |
| P005               | 28                   | 33.5000                   |
| P006               | 50                   | 20.0000                   |

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4. Return to the worksheet, Right-click the **Number of Units Sold** filter on the **Filters** shelf and select **Add to Context**.



5. The filter pill changes to **gray**, indicating that it has become a **Context Filter**.

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